

2018556059

Colist

Barker

Subject:

Date: / /

Barker

$$151) a) f(x_1) = (1+2) - (3+4) - (5+6) - (7+8) = -30$$

$$f(x_2) = (2+3) - (4+5) - (6+7) - (8+1) = -26$$

$$f(x_3) = (4+5) - (6+3) - (2+2) - (8+1) = -18$$

$$f(x_4) = (5+3) - (6+1) - (2+2) - (8+4) = -20$$

$$f(x_5) = (2+3) - (5+2) - (1+2) - (8+5) = -18$$

The order is $\boxed{x_3, x_4, x_5, x_2, x_1}$

b) —

$$10 c) i) x_3 = 4563 / 7281$$

$$x_5 = 5361 / 7284$$

→

$$a_1 = 4563 \ 7284$$

$$a_2 = 5361 \ 7281$$

$$ii) x_5 = 239 / 2128 / 5$$

$$x_4 = 536 / 1728 / 4$$

→

$$a_3 = 239 \ 1728 \ 5$$

$$a_4 = 536 \ 2128 \ 4$$

$$2 - z = \frac{1}{1+e^{-z}}$$

a)

$$\text{Node 1) } z_1 = W_{11} * x_1 + W_{21} * x_2 - W_1 = 2$$

$$\text{Node 2) } z_2 = W_{22} * x_2 + W_{12} * x_1 - W_2 = 2$$

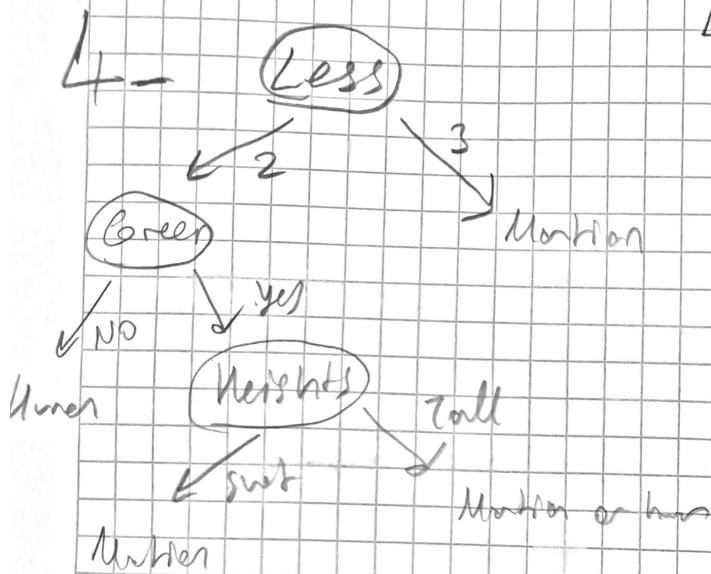
$$\text{Node 3) } z_3 = W_{13} * y_1 + W_{23} * y_2 - W_3 = -1.16$$

$$y_1) n(2) = -0.88$$

$$y_2) n(2) = -0.88$$

$$y_3) n(-1.16) = -0.24$$

15 3-1-No, 2-YES, 3-No, 4-YES, 5-NO



Markhan

(Less=3) V (Less=2 A Green=Yes A

Height=Short) V (Less=2 A

Green=Yes A Height=tall)

Huan?

(Less=2 A Green=No) V

(Less=2 A Green=Yes A

Height=tall)

Information gain = entropy(parent) - [weighted average] * entropy(children) = Entropy(S)

entropy = $\sum_i -p_i \log_2(p_i)$

$\sum((|S_v|/|S|))$

Entropy(green) = $-p_N \log_2 p_N - p_U \log_2 p_U$

$= -\frac{5}{10} \log_2 \frac{5}{10} - \frac{5}{10} \log_2 \frac{5}{10} = 1$

Entropy(Sv)

Entropy(Less) = $-p_3 \log_2 p_3 - p_2 \log_2 p_2 = 0.881$

Entropy(Species) = $-p_M \log_2 p_M$

Entropy(Height) = $-p_8 \log_2 p_8 - p_7 \log_2 p_7 = 0.222$

$p_U \log_2 p_U$

2) b) $W_x' = W_x - a \left(\frac{\partial \text{Error}}{\partial W_x} \right)$, assume (a) learning rate is 1

$W_3' = W_3 - 1 * n(z_2) * (1 + n(z_2)) = 0.5 - 1 * (-0.24) * (1 - 0.24)$

$= 0.6824$

$W_{13}' = W_{13} - 1 * n(W_{13}) * (1 + n(W_{13})) = 0.5 - 1 * (-0.62) * (1 - 0.62)$

$= 0.7356$

$W_{11}' = W_{11} - 1 * n(W_{11}) * (1 + n(W_{11})) = 1 - 1 * (-0.73) * (1 - 0.73)$

$= 0.6971$